Software Engineering: Requirements Engineering & UML Modelling
Problem Statement #27 | Campus & Academic Operations: Smart Cities, Transport & Logistics Warehouse Inventory & Pallet Location Tracker

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3. Core Use-Case Flow Specification

Field	**Specification**
Use Case Name & ID	**UC-01: Record Pallet Storage**
Goal	To enable a Warehouse Operator to scan a pallet, assign it to a valid 3D shelf location, verify that the rack can safely support the pallet's weight, and record the storage operation for future tracking and retrieval.
Primary Actor	**Warehouse Operator**
Secondary Actors	**Logistics Supervisor, Barcode/RFID Scanner**
Preconditions	1. Warehouse Operator is authenticated and logged into the warehouse management system.2. The pallet has a valid barcode or RFID tag registered in the system.3. The proposed rack, level, and slot are available for pallet placement.4. Rack maximum weight capacity is available in the system.
Postconditions	1. The pallet is assigned a valid 3D warehouse coordinate (aisle, rack, level, slot).2. The system records the pallet's storage location and weight.3. A time-stamped storage movement record is created.4. The pallet becomes searchable using its SKU or pallet ID.
Main Success Scenario (MSS)	1. Warehouse Operator selects **'Store Pallet'**.2. Operator scans the pallet's barcode/RFID tag **(UC-1.1 «include»)**.3. System retrieves the pallet ID, SKU, and weight from the inventory database.4. Operator enters or selects the proposed aisle, rack, level, and slot coordinates.5. System validates that the selected 3D location is available and checks the rack's remaining load capacity **(UC-1.2 «include»)**.6. System confirms that the pallet weight does not exceed the rack's maximum structural limit.7. System records the pallet location and creates a time-stamped stock movement record.8. System displays **'Pallet Stored Successfully'** along with the pallet ID and assigned 3D coordinates.
Alternate Flows	**5a. Rack Weight Capacity Exceeded:**1. System calculates that adding the pallet would exceed the rack's maximum load capacity.2. System rejects

Field	Specification
	the proposed placement and displays an overload warning.3. Operator selects another valid rack/location.4. Flow resumes at Step 4 of the Main Success Scenario.**4a. Location Unavailable:**1. System detects that the selected slot is already occupied or invalid.2. System displays **'Selected location unavailable.'**3. Operator selects another available location.4. Flow resumes at Step 4 of the Main Success Scenario.**2a. Invalid Barcode/RFID:**1. System cannot identify the scanned pallet.2. System displays **'Invalid or unregistered pallet.'**3. Operator rescans the pallet or reports the issue to the Logistics Supervisor.4. Flow terminates until a valid pallet is identified.
Special Requirements	1. The pallet location lookup and rack-capacity validation should complete within **100 ms** under the specified 100,000-bin workload.2. All successful storage and movement records shall contain a timestamp and operator identification.3. Only authorized Warehouse Operators and Logistics Supervisors shall be permitted to perform their respective inventory operations.

4. UML Use-Case Diagram Specification

The UML Use-Case Model illustrates interactions between key actors and the system boundary of the **Warehouse Inventory & Pallet Location Tracker**. It explicitly includes the required **«include»** and **«extend»** relationships:

- **Actors:** Warehouse Staff (Primary), Warehouse Manager (Primary), System / Notification Service (Secondary).
- **Primary Use Cases:** Register Inventory, Track Inventory Status, Assign Pallet Location, Monitor Stock Levels & Overdue Movements, Verify & Confirm Inventory Update.
- **«include» Relationship:** UC1 (Register Inventory) *«include»* UC-Inc (Validate Product & Pallet Details). [Mandatory execution during inventory registration]
- **«extend» Relationship:** UC-Ext (Attach Product Image / Supporting Document) *«extend»* UC1 (Register Inventory). [Optional execution triggered when the warehouse staff has supporting evidence]
- **«extend» Relationship (Stock Alert):** UC-Ext2 (Trigger Low-Stock Alert) *«extend»* UC4 (Monitor Stock Levels & Overdue Movements) on condition [Stock Level < Minimum Threshold].